Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details



a. Team name: Kakwe (ကာကွယ်)
b. Team leader name: Ricky
c. Problem Statement: Recruitment ads on Facebook, Viber and Telegram lure young job seekers from Myanmar and the region into forced online scam work; OHCHR says at least 120,000 people in Myanmar may be held in it. They need to see the warning signs, in their own language, before they pay, send documents or travel.

Brief about the idea

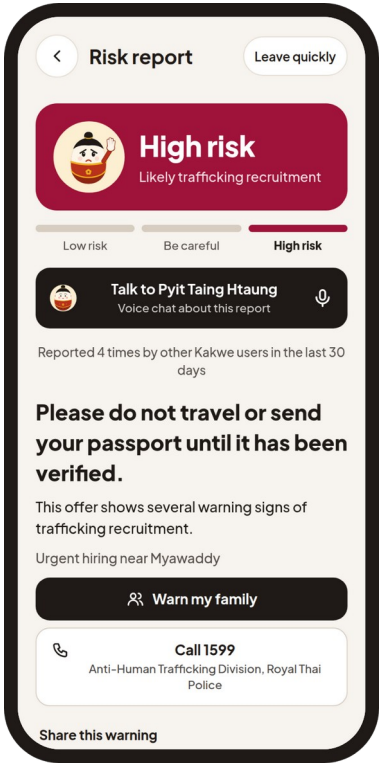
Kakwe checks a job offer before you pay, send documents or travel.

Kakwe (ကာကွယ်, Burmese for "protect") is an Android app and a web demo for migrant job seekers from Myanmar, Thailand, Indonesia and the Philippines.

Paste a job ad, chat, link or screenshot, or tap the floating doll over any Android app. Gemini reads the evidence, Google Search checks named companies, and fixed rules set the verdict: Low risk, Be careful or High risk.

The report quotes the user's own words, gives a verified hotline and speaks the warning in English, Burmese, Bahasa Indonesia, Tagalog or Thai.

The package name and web addresses keep the earlier project name saferoute.





Opportunities

How different is it from any of the other existing ideas?

It reads whole chat screenshots in Burmese and four other languages, not only links or numbers. Job Shield puts the job ad next to the recruiter's chat and shows each gap. Every warning quotes the words behind it.

How will it be able to solve the problem?

It works at the moment of risk: before paying, sending a passport or travelling. One tap on the floating doll checks any Android app. The report gives the next step, a verified hotline and a way to warn family.

USP of the proposed solution

Gemini finds evidence; deterministic rules decide. Wording stays in likelihood, never accusation. A Burmese voice companion answers from the finished report. The benchmark and its dataset are public.

List of features offered by the solution

Check anything
Paste a message or link, or share a screenshot. Get Low risk, Be careful or High risk, with the quoted words.

Job Shield
Compares the job ad with what the recruiter asks in chat and shows each gap.

Screen bubble
Android. A floating doll checks whatever is on screen, in any app, with one tap.

Message Guard
Android, opt-in. Watches the chat apps you choose and flags risky messages.

Voice companion
Talk to Pyit Taing Htaung about a report, in your language, on Gemini Live.

Warn my family
Sends the warning to a trusted contact by SMS or the share sheet.

Scam network
Opt-in. Repeat numbers and handles, kept as keyed hashes, raise the next warning.

Five languages
English, Burmese, Bahasa Indonesia, Tagalog and Thai. Leave quickly clears the screen.

Process flow diagram or Use-case diagram

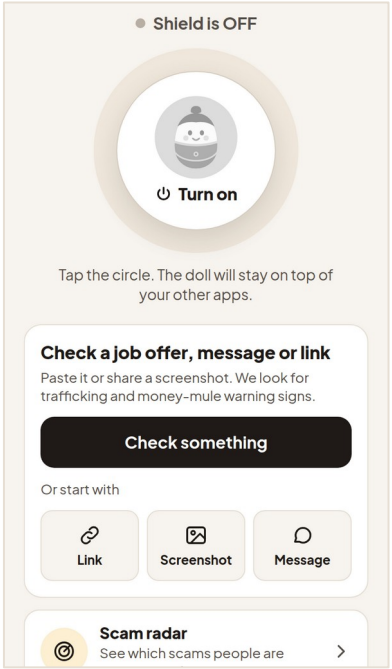
Ma Su's journey (a fictional story)

A Viber message offers a factory job in Thailand → She pastes it into Check, or taps the floating doll → High risk. Her own words are quoted as warning signs → Job Shield: the ad says no fees, the chat asks for a deposit → She asks Pyit Taing Htaung, in Burmese → She warns her mother. She does not pay or travel

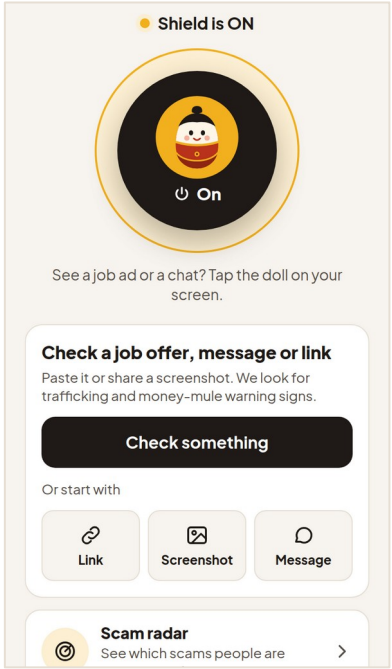
The check flow, within a 15 second budget

Text, link or screenshot, with the user's language → Gemini extraction: flags, quotes and links as JSON → In parallel: Search grounding, link checks, scam network → Likelihood guard rewrites certainty words → Rules engine sets the score and the verdict → Report, spoken warning (TTS) and voice chat (Live)

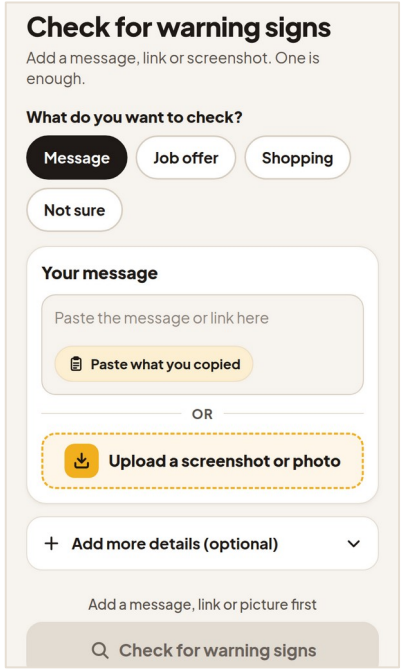
Wireframes/Mock diagrams of the proposed solution (optional)



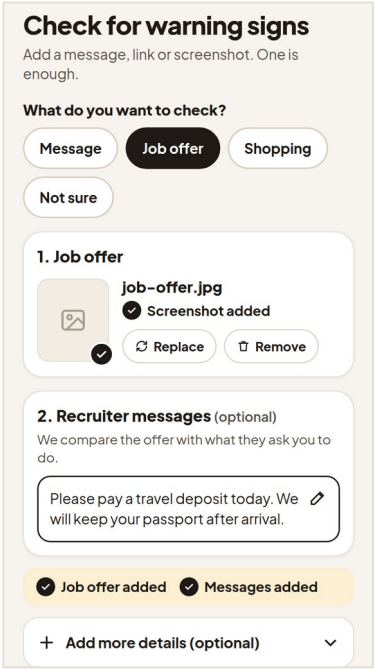
Home, shield off



Home, shield on



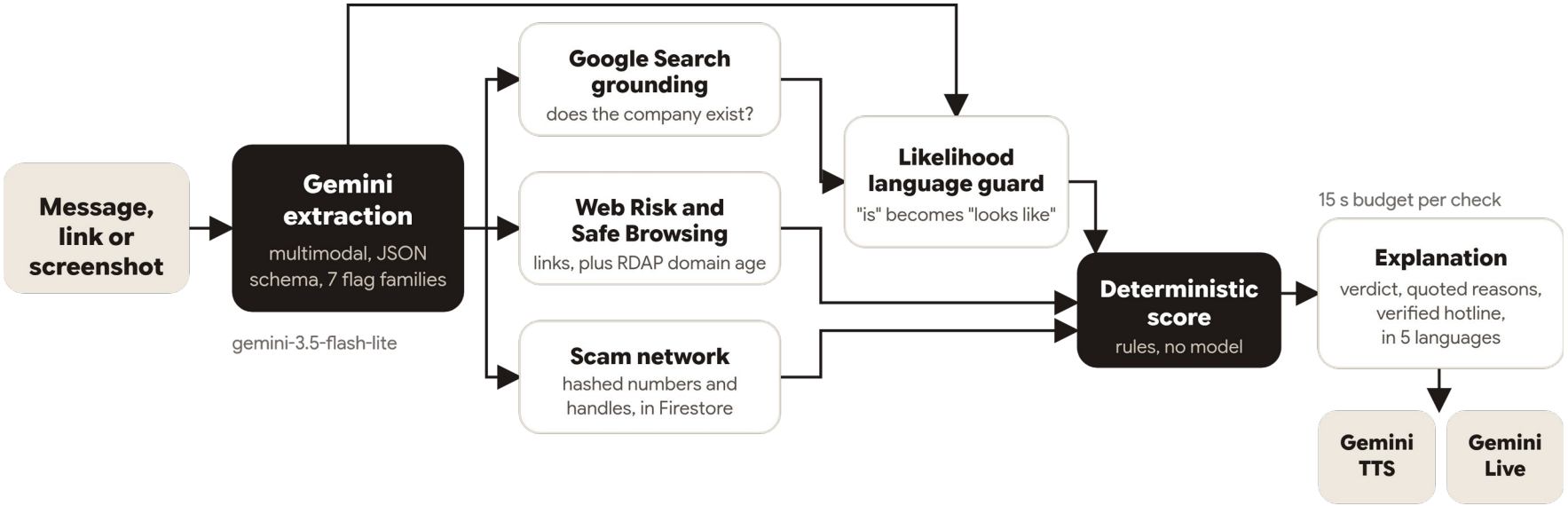
Check, message



Check, job offer filled

Approved design mockups for Home (one big shield button) and the Check v3 page with kind chips, in English. The logo row and the footer line are cropped.

Architecture diagram of the proposed solution



Hosting

API on Cloud Run (asia-southeast1, Singapore). Firebase for sign-in, data, web hosting and test builds.

Technologies to be used in the solution

Technology	Use in Kakwe
Gemini API, gemini-3.5-flash-lite	Multimodal extraction with a JSON response schema; Job Shield fact extraction
Grounding with Google Search	Checks that named agencies and companies exist
Gemini TTS, gemini-3.8-flash-tts	Spoken warnings in five languages
Gemini Live API, gemini-3.8-live	The voice companion, Pyit Taing Htaung
Web Risk API, Safe Browsing API	Known malicious links
Firebase Auth, Firestore	Anonymous and Google sign-in, synced history and contacts, the scam network
Firebase Hosting, App Distribution	The web demo and Android test builds
Cloud Run, asia-southeast1	Hosts the check API and the voice relay
Expo 52, React Native 0.76, TypeScript	The Android app and the web demo
Hono on Node 22, TypeScript	The check API, the rules engine and the voice relay
Kotlin Android module	The screen bubble and Message Guard

Estimated implementation cost (optional)

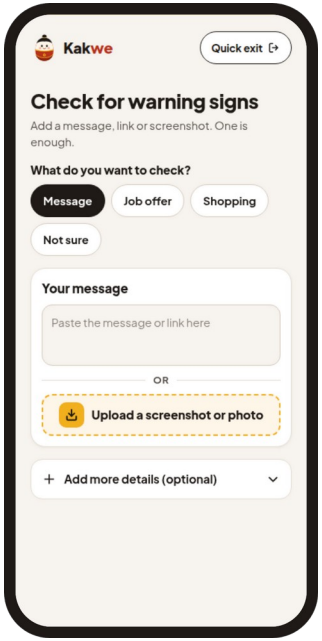
Model calls per check

- One Gemini extraction call (gemini-3.5-flash-lite), text plus an optional image.
- A second Gemini call with Google Search grounding, only when a company, agency or website is named.
- One Web Risk or Safe Browsing lookup and one RDAP lookup per link.
- Gemini TTS only when a spoken warning is played.
- A voice session lasts at most 120 seconds and counts as two checks.

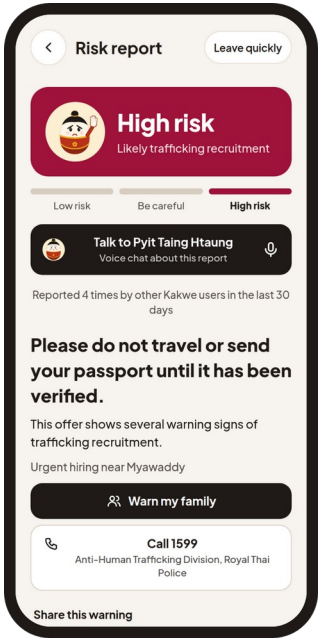
Cost controls in the API

- An identical check within 10 minutes comes from the result cache.
- Hourly limits: 20 checks anonymous, 60 signed in, and 30 requests a minute per IP.
- Each check has a 15 second budget; a late stage returns "unknown".
- The Firebase project serves the web demo with no billing account linked.
- We have not measured a cost per check yet, so this deck gives no dollar figure.

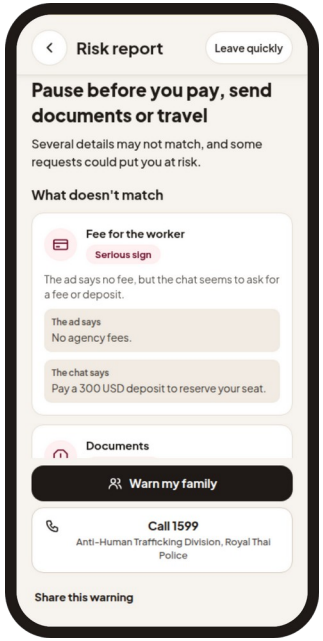
Snapshots of the prototype



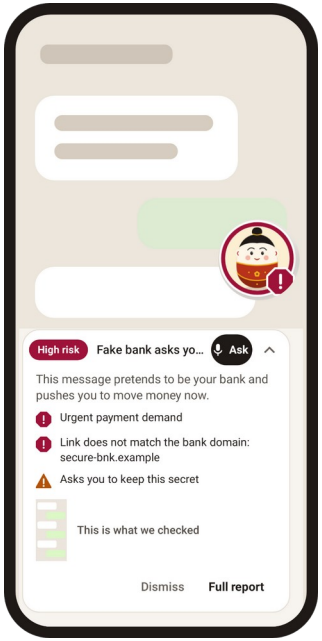
Check: pick what to check



High risk report



Job Shield: what doesn't match



Screen bubble (Android)



Burmese report, voice and family

Web build captures with fictional inputs and the built-in demo data. The bubble shows the Android overlay views on a drawn chat.

Prototype Performance report/Benchmarking

31/31

Text eval cases correct

0 responses with certainty wording after the guard. p50 2.8 s, p95 4.4 s.

89.7

Protection Score, 24 synthetic screenshots

5 languages, 0 critical failures, 48/48 quotes found in the image.

about 67

Protection Score, 96 real screenshots

Provisional. Draft labels from two models; no human review yet.

10/10

Job Shield ad and chat pairs

No legit pair reached Pause. No scam pair stayed at the lowest tier.

93 + 28

Public dataset on Hugging Face

Real redacted and synthetic screenshots with labels.

98%

Verdict stability, 210 variants

Low resolution, JPEG, crop, rotation, dark mode, overlays and prompt injection.

gemini-3.5-flash-lite, production prompt, runs of 16 October 2026. A live link check also flags Google's Safe Browsing test phishing URL as High risk.

Additional Details/Future Development (if any)

Next

1. **Native review of every Burmese string**
2. **Human review of the real-screenshot labels**
3. **More languages for the region**
4. **Pilots with migrant worker NGOs**

Known limits today

- Burmese app strings are machine translated and not yet reviewed by a native speaker.
- Real-screenshot labels are drafts from two models, so that score is provisional.
- The web demo has no screen bubble or Message Guard; both need Android.

Provide links to your:

- GitHub Public Repository: https://github.com/minthein128/scamshield
- Demo Video Link (3 Minutes)
- Final Product Link: web demo https://saferoute-aibc.web.app, Android APK https://saferoute-aibc.web.app/saferoute.apk

Public dataset: https://huggingface.co/datasets/Rickaym/saferoute-scam-screenshots

Team: Ricky (team leader), Min Thein Kyaw

Contact: support@akkhayar.com



Safety and privacy

Likelihood words

"Shows signs of", never "is a scam". A guard rewrites certainty phrases in all five languages.

No accusations

No named person, company, number or handle is called a criminal.

Consent first

Signal sharing and Message Guard stay off until the user turns them on.

Hashed indicators

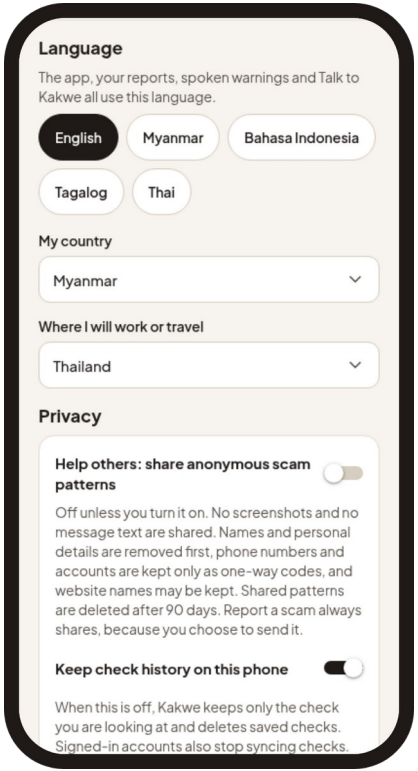
Phone numbers and handles are kept as one-way hashes; signals expire after 90 days.

Leave quickly

One button on every report clears the screen back to Home and stops audio.

Slim sync, 18+

History keeps the verdict, a score band, the title and the summary, and can be turned off.



Images and message text are not stored or logged. Personal details are redacted before a signal is kept.

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Thank you

